

Official Agricultural Extension Advisory: Brown Spot in Rice (Paddy)

Causal Pathogen: Bipolaris oryzae (teleomorph Cochliobolus miyabeanus)

Issuing Source: ICAR-NRRI & IRRI Rice Health Management Guide | Date: 2024-03-22

1. Identification and Morphological Symptoms

Affects seedlings, leaves, leaf sheaths, and glumes. On leaves, spots are circular to oval, resembling sesame seeds (sesame spot). Small spots are dark brown or purplish-brown; fully developed spots are circular to oval with a light brown or gray center and a distinct dark reddish-brown border, frequently surrounded by a yellow halo. Heavily infected leaves turn yellow and wither prematurely. Infected grains develop dark brown or black discoloration, causing poor germination and grain spotting (pecky rice).

2. Environmental Epidemiology & Causes

Historically associated with the Great Bengal Famine of 1943. Associated with nutrient-deficient soils, especially potassium, silicon, manganese, and iron deficiency. Prevalent in upland, drought-stressed, sandy, or poorly drained soils with temperatures of 25-30 deg C and high humidity (>80%).

3. Cultural Management and Field Prevention

Prevention: Ensure balanced plant nutrition by soil testing; correct nutrient deficiencies especially potassium, zinc, and silicon. Ensure uniform seedbed irrigation and avoid severe moisture stress.

Cultural Practices: Apply well-decomposed organic manure or compost @ 5 tonnes/ha. Practice proper water management to avoid soil drying during vegetative stages. Apply potash in two split applications (50% basal + 50% panicle initiation).

4. Organic and Biological Management

Hot water seed treatment at 52-54 deg C for 10-15 minutes to eliminate internal seed-borne mycelium. Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg seed. Foliar spray of fresh neem seed kernel extract (NSKE 5%) or garlic bulb extract @ 2% at disease onset.

5. Verified Chemical Interventions

Seed treatment with Carbendazim 12% + Mancozeb 63% WP @ 2.0 g/kg seed. Foliar spray of Mancozeb 75% WP @ 2.0 g/L or Propiconazole 25% EC @ 1.0 mL/L or Tebuconazole 25.9% EC @ 1.5 mL/L at the boot leaf stage.

• **Active Ingredient:** Propiconazole 25% EC | **Reference Formulation:** Tilt (standard reference) | **Registrant:** Syngenta India Ltd | **Source:** CIBRC Registered Fungicide Gazette (2024-01-18)

• **Active Ingredient:** Carbendazim 12% + Mancozeb 63% WP | **Reference Formulation:** Saaf | **Registrant:** UPL Ltd | **Source:** ICAR-NRRI Advisory (2024-02-15)

Notice: Chemical recommendations are derived strictly from central agricultural research advisories. Farmers must consult local district agricultural extension officers for local dosage and label claims.